



SHIVANSH GARG



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63
2024	B.E in Computer Science and Engineering	UIET, Panjab University	9.37
2020	CBSE	Shaheed Bhai Dila Ji Public School	94.2%
2018	CBSE	Sacred Heart Convent School	97.8%

QUALIFYING EXAMS

- **Graduate Aptitude Test in Engineering (GATE) Rank: 75**

SCHOLASTIC ACHIEVEMENTS

- **Department Rank 3:** Current rank in M.Tech (Computer Science & Engineering), IIT Delhi.

IIT DELHI THESIS

Title: Atrial Fibrillation Detection Using PPG Signals in Wearable Devices

Supervisor: Prof. Rahul Garg

Description: Developing **machine learning** algorithms to enhance **Atrial Fibrillation detection** using **PPG signals from wearable devices**. This research focuses on **minimizing false positives** by distinguishing cardiac events from motion artifacts and investigating PPG-derived features to identify **subtypes of AF**, thereby improving remote cardiac monitoring precision.

PROJECTS

- **P-Wave Detection in Electrocardiograms (Prof. Rahul Garg) *Minor Project* :**
 - Performed a **systematic literature review** of **30** studies on automated P-wave detection covering signal processing, **machine learning**, and **deep learning approaches**.
 - **Implemented and validated algorithms** and benchmarked the **NeuroKit2** library, assessing overall **reproducibility**.
 - Conducted **comparative evaluation** of detection methods on **ECG databases**, analyzing performance and limitations.
- **Visual Question Answering with Multi-Modal Transformers (Prof. Parag Singla):**
 - Built a multi-modal transformer with a **ResNet image encoder** and **transformer text encoder**.
 - Used **cross-attention** fusion to improve **visual-text reasoning** on the CLEVR dataset.
 - Improved performance using **BERT embeddings and focal loss**, and validated generalization through **zero-shot**.
- **LaTeX to Markdown Converter: (Prof. Smruti Ranjan Sarangi):**
 - Created a compiler using **Flex** for lexical analysis, **Bison** for parsing, and **AST traversal** for Markdown generation.
- **Neural Cleanse: Backdoor Attack Detection in Deep Neural Networks (Prof. Huzur Saran):**
 - Implemented and evaluated **blind backdoor attacks** on **deep neural networks** to study stealthy adversarial behavior.
 - Implemented corresponding **defense mechanisms** using **differential privacy** and **certified robustness**, and analyzed their effectiveness in mitigating attacks.
- **Custom Signal Handling and Scheduler Enhancement in xv6 (Prof. Smruti Ranjan Sarangi):**
 - Developed custom signal handling to support process termination, suspension, resumption, and user-defined actions.
 - Enhanced the **xv6 scheduler** with profiling metrics and a **priority boosting mechanism** to ensure fairness.
- **Search-Based Agent for ASR Error Correction (Prof. Rohan Paul):**
 - Developed a **local search agent** to correct phonetic character substitutions and missing word errors in text transcribed by **speech recognition systems**.

TECHNICAL SKILLS

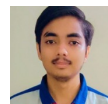
- **Programming Languages :** C/C++, Python, SQL
- **Software and Tools :** PyTorch, scikit-learn, NeuroKit2, NumPy, Pandas, xv6, Linux Shell, Git, Unity.

TEACHING ASSISTANTSHIP

- Teaching Assistant, **Principles of Artificial Intelligence (COL333)** | Prof. Mausam | Diwali 2025
- Teaching Assistant, **Analysis and Design of Algorithms (COL351)** | Prof. Keerti Chaudary | Holi 2025
- Teaching Assistant, **Analysis and Design of Algorithms (COL351)** | Prof. Rohit Vaish | Diwali 2024



SHIVANSH GARG



IIT COURSE

Degree	Institute	CGPA	Dept. Rank
M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63	---

COURSES DONE

Advanced Data Structures, Artificial Intelligence, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Teaching Practicum In Computer Science And Engineering, Machine Learning, Resource Management In Computer Systems, Networks & System Security, Minor Project, Teaching Practicum In Computer Science And Engineering